

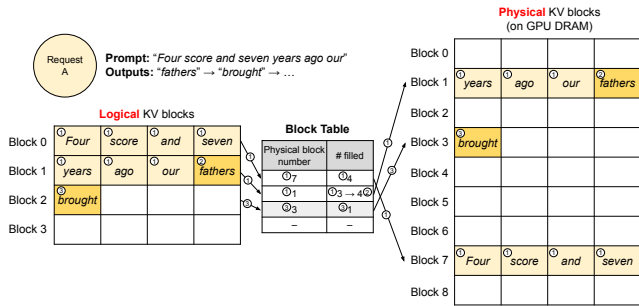


Figure 6. Block table translation in vLLM.

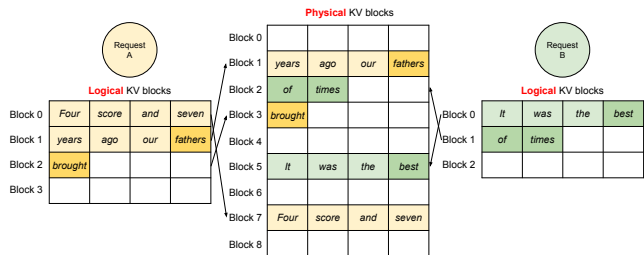


Figure 7. Storing the KV cache of two requests at the same time in vLLM.

divides it into *physical KV blocks* (this is also done on CPU RAM for swapping; see §4.5). The *KV block manager* also maintains *block tables*—the mapping between logical and physical KV blocks of each request. Each block table entry records the corresponding physical blocks of a logical block and the number of filled positions. Separating logical and physical KV blocks allows vLLM to dynamically grow the

requests and the latest tokens for generation phase requests) as one sequence and feeds it into the LLM. During LLM's computation, vLLM uses the PagedAttention kernel to access the previous KV cache stored in the form of logical KV blocks and saves the newly generated KV cache into the physical KV blocks. Storing multiple tokens within a KV block (block size > 1) enables the PagedAttention kernel to process the